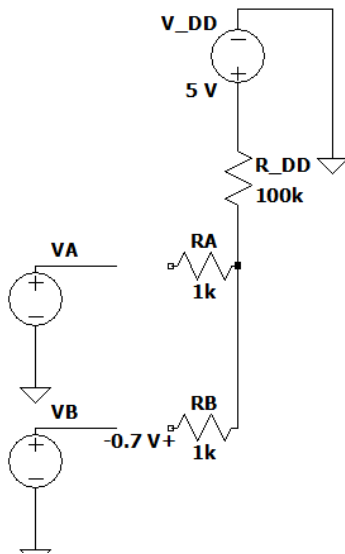


Exercise 1

Solution:

- a) Assuming High input = $V_{DD} = 5\text{ V}$ & Low input = 0 V , we assume, D_A will be off & D_B will be on for (1,0).



KCL gives, $I_{DD} = I_{DA} + I_{DB}$

$$\rightarrow \frac{5-V_O}{R_{DD}} = 0 + \frac{V_O-(V_B+0.7)}{R_B}$$

$$\rightarrow \frac{5-V_O}{100} = \frac{V_O-0-0.7}{1} \rightarrow V_O = 0.743\text{ V}$$

Verification: Voltage across D_A , $V_{D_A} = (V_O - I_A \times R_A - V_A)$

$$= (0.743 - 0 - 5) = -4.267 < 0.6\text{ V}$$

Voltage across D_B , $V_{D_B} = 0.7$

Also, the output voltage is lower than that of the high input.

- b) For (0,1), D_A will be on & D_B will be off. However, since R_A & R_B are same, V_O will be the same. Now,

$$I_{DD} = I_{DA} + I_{DB} = \frac{V_O - V_A - 0.7}{R_A} + 0 = \frac{0.743 - 0 - 0.7}{1} = 0.043\text{ mA}$$

Note: This is also the current that is flowing through R_A .

- c) Both diodes will be on in this case.

$$\text{So, } I_{DD} = I_{DA} + I_{DB} = \frac{V_O - V_A - 0.7}{R_A} + \frac{V_O - V_B - 0.7}{R_B}$$

$$\rightarrow \frac{V_{DD} - V_O}{R_{DD}} = \frac{V_O - 0 - 0.7}{1} + \frac{V_O - 0 - 0.7}{2} \rightarrow \frac{5 - V_O}{100} = \frac{3(V_O - 0.7)}{2} \rightarrow V_O = 0.728\text{ V}$$

$$\therefore I_{DA} = \frac{V_o - 0.7}{R_A} = \frac{0.728 - 0.7}{1} = 0.028 \text{ mA}$$

$$\therefore I_{DB} = \frac{V_o - 0.7}{R_B} = \frac{0.7228 - 0.7}{2} = 0.014 \text{ mA}$$

d) Voltage of the node between D_B & $R_B = V_B + 0.7 = 0 + 0.7 = 0.7 \text{ V}$

Exercise 2

Solution:

From **Exercise 1**, we found that the output voltage for (1,0) & (0,1) is 0.743V. The other case when the output voltage will be low is (0,0). For this, KCL gives,

$$I_{DD} = I_{DA} + I_{DB} \rightarrow \frac{V_{DD} - V_o}{R_{DD}} = \frac{V_o - V_A - 0.7}{R_A} + \frac{V_o - V_B - 0.7}{R_B}$$

$$\rightarrow \frac{5 - V_o}{100} = \frac{V_o - 0.7}{1} + \frac{V_o - 0.7}{1} \rightarrow V_o = 0.721 \text{ V}$$

Thus the lower threshold voltage of output for the AND gate =

$$\max(V_{o_{0,0}}, V_{o_{0,1}}, V_{o_{1,0}})$$

$$= \max(0.721, 0.743, 0.743) = 0.743 \text{ V}$$

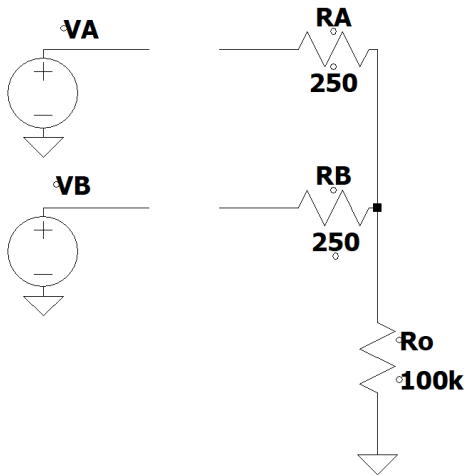
Exercise 3

Solution:

Step 1:

Case	DA	DB
0,0	Off	Off
0,1	Off	On
1,0	On	Off
1,1	On	On

(0,0):



Step 2:

$$I_{DA} = I_{DB} = 0$$

$$\therefore I_o = I_{DA} + I_{DB} = 0$$

$$V_A = V_B = 0.2 \text{ V}$$

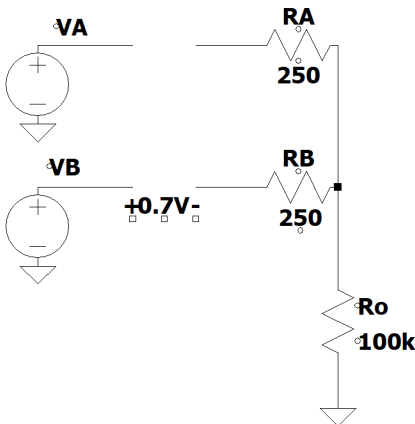
Both I_{DA} & I_{DB} flows from their respective sources to the ground.

Step 3:

$$P = (V_A - 0) \times I_{DA} + (V_B - 0) \times I_{DB}$$

$$\therefore P_{0,0} = (0.2 - 0) \times 0 + (0.2 - 0) \times 0 = 0 \text{ W}$$

(0,1):



Step 2:

$$I_{DA} = 0, I_{DB} = \frac{V_B - 0.7}{250 + 100 \times 10^3} = \frac{5 - 0.7}{100250} = 42.9 \mu\text{A}$$

$$V_A = 0.2 \text{ V}, V_B = 5 \text{ V}$$

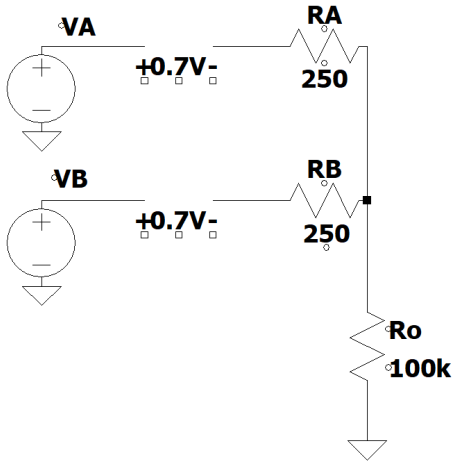
Step 3:

$$P = (V_A - 0) \times I_{DA} + (V_B - 0) \times I_{DB}$$

$$\therefore P_{0,1} = (0.2 - 0) \times 0 + (5 - 0) \times 42.9 = 214.5 \mu\text{W}$$

(1,0) can be found similarly. Note that both the diode branches have the same resistances & input voltages. Hence, the result for (1,0) will be the same as (0,1). If those were not the same, the results would differ.

(1,1):



Step 2:

$$I_{DA} = I_{DB} = \frac{V_A - 0.7 - V_o}{250}, I_o = I_{DA} + I_{DB} = \frac{V_o - 0}{100 \times 10^3}$$

$$\rightarrow 2 \times \frac{V_A - 0.7 - V_o}{250} = \frac{V_o - 0}{100 \times 10^3} \rightarrow V_o = 4.29 V$$

$$\therefore I_{DA} = I_{DB} = \frac{V_o}{100 \times 2} = 21.45 \mu A$$

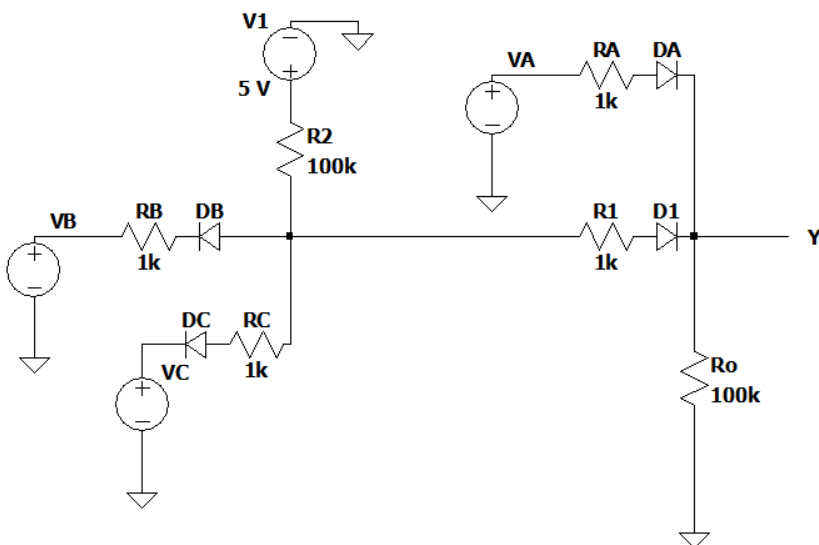
$$V_A = V_B = 5 V$$

Step 3:

$$P_{1,1} = (5 - 0) \times 21.45 + (5 - 0) \times 21.45 = 214.5 \mu W$$

Exercise 4

Solution:



a) $Y = A + BC$

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

b) Output voltage is high for the last 5 rows of input in the truth table.

Case (0,1,1):

D_A, D_B & D_C all are off, D_1 on

Hence, KCL at the output voltage node of the AND gate, $V_{o(AND)}$ gives,

$$I_{R2} = I_{DB} + I_{DC} + I_{R1} \rightarrow \frac{5-V_{o(AND)}}{100} = 0 + 0 + \frac{V_{o(AND)}-V_Y-0.7}{1}$$

$$\rightarrow 101 V_{o(AND)} - 100V_Y = 75$$

KCL at Y gives,

$$I_{R1} + I_A = I_{R0} \rightarrow \frac{V_{o(AND)}-V_Y-0.7}{1} = \frac{V_Y-0}{100} \rightarrow 100V_{o(AND)} - 101V_Y = 70$$

Solving the two equations, $V_{o(AND)} = 2.86 V, V_Y = 2.14 V$

Case (1,0,0):

D_A, D_B & D_C all are on, D_1 can be assumed off (since output of AND gate is low)

KCL at AND gate output node,

$$I_{R2} = I_{DB} + I_{DC} \rightarrow \frac{5-V_{o(AND)}}{100} = \frac{V_{o(AND)}-0-0.7}{1} \times 2 \rightarrow V_{o(AND)} = 0.721 V$$

KCL at Y,

$$I_{RA} = I_{R0} \rightarrow \frac{5-(V_Y+0.7)}{1} = \frac{V_Y-0}{100} \rightarrow V_Y = 4.25 V$$

Case (1,0,1):

D_A, D_B on, D_C off, D_1 can be assumed off (since output of AND gate is low)

KCL at AND gate output node,

$$I_{R2} = I_{DB} + I_{DC} \rightarrow \frac{5-V_{o(AND)}}{100} = \frac{V_{o(AND)}-0-0.7}{1} \rightarrow V_{o(AND)} = 0.742 V$$

KCL at Y,

$$I_{RA} = I_{R0} \rightarrow \frac{5-V_Y-0.7}{1} = \frac{V_Y-0}{100} \rightarrow V_Y = 4.25 V$$

Case (1,1,0): Same as Case (1,0,1)

Case (1,1,1):

D_A, D_1 on, all other diodes are off

KCL at Y,

$$I_{RA} + I_{R1} = I_{R0} \rightarrow \frac{5-V_Y-0.7}{1} + \frac{5-V_Y-0.7}{101} = \frac{V_Y-0}{100} \rightarrow V_Y = 4.25 V$$

$$\therefore V_{o(AND)} = 5 - I_{R1} \times 100 = 5 - \frac{5-4.94-0.7}{101} \times 100 = 4.96 V$$

Thus, the higher threshold of output voltage =
 $\min(V_Y \text{ of all high output cases}) = 2.14 V$

Output voltage is low for the first 3 rows of input in the truth table.

Case (0,0,0):

D_A off, D_B & D_C on, D_1 can be assumed off

(since the output of the AND gate is low)

KCL at AND gate output node,

$$I_{R2} = I_{DB} + I_{DC} \rightarrow \frac{5-V_{o(AND)}}{100} = \frac{V_{o(AND)}-0-0.7}{1} \times 2 \rightarrow V_{o(AND)} = 0.721 V$$

$V_Y = 0$ (since all diode branches connected to Y are open)

Case (0,0,1):

D_A off, D_B on, D_C off, D_1 can be assumed off

KCL at AND gate output node,

$$I_{R2} = I_{DB} + I_{DC} \rightarrow \frac{5-V_{o(AND)}}{100} = \frac{V_{o(AND)}-0-0.7}{1} \rightarrow V_{o(AND)} = 0.742 V$$

$V_Y = 0$ (for the same reason as previous case)

Case (0,1,0): Same as Case (0,0,1)

Thus, lower threshold of output voltage = $\max(V_Y \text{ of all low output cases})$
 $= 0 \text{ V}$

c) Power dissipation of all cases:

Case (0,0,0):

$$P_{0,0,0} = (5 - 0) \times I_{R2} = 5 \times \frac{5-0.721}{100} = 0.214 \text{ mW}$$

Case (0,0,1):

$$P_{0,0,1} = (5 - 0) \times I_{R2} = 5 \times \frac{5-0.742}{100} = 0.213 \text{ mW}$$

Case (0,1,0): Same as Case (0,0,1).

Case (0,1,1):

$$P_{0,1,1} = (5 - 0) \times I_{R2} = 5 \times \frac{5-2.86}{100} = 0.107 \text{ mW}$$

Case (1,0,0):

$$P_{1,0,0} = (5 - 0) \times I_{R2} + (5 - 0) \times I_{RA} = 5 \times \left(\frac{5-0.721}{100} + \frac{5-4.95}{1} \right) = 0.464 \text{ mW}$$

Case (1,0,1):

$$P_{1,0,1} = (5 - 0) \times I_{R2} + (5 - 0) \times I_{RA} = 5 \times \left(\frac{5-0.742}{100} + \frac{5-4.25-0.7}{1} \right) = 0.463 \text{ mW}$$

Case (1,1,0): Same as Case (1,1,0).

Case (1,1,1):

$$P_{1,1,1} = (5 - 0) \times I_{R1} + (5 - 0) \times I_{RA} = 5 \times \left(\frac{5-4.25-0.7}{101} + \frac{5-4.25-0.7}{1} \right) = 0.252 \text{ mW}$$

$\therefore$ Maximum Power Dissipation, $P_{max} = 0.464 \text{ mW}$

$\therefore$ Average Power Dissipation, $P_{avg} = \frac{0.214+0.213 \times 2+0.464+0.107+0.463 \times 2+0.252}{8}$
 $= 0.299 \text{ mW}$